Reading Directions

Not a syllabus. Questions to bring to specific sources.

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Contents

The programme's questions

The programme reconstructs structure from relations, not spaces. Point-spaces (the real line, Stone spaces, Hilbert spaces) are outcomes of the reconstruction, not its starting material. This is a bet: that the relational description — contexts, reference frames, directed refinement — is the right starting point, and that the structures usually assumed (points, σ-algebras, global states) are *reconstructed* from it.

Three central questions organize the reading:

1. **What must be added to observation to get structure?**

   The persistent question across all phases:

   - Phase 1: observation → dynamics (semigroup)
   - Phase 2–3: observation → geometry (curvature, metric, action)
   - Phase 4–6: observation → probability (CE), realization (descent), value-definiteness (distributivity)

   The reading form: *Where does someone identify the exact condition that forces a valuation to behave — and what structure does that condition live in?*

2. **Reconstruction without points.**

   When can observation be modeled relationally — via contexts and reference frames — rather than by presupposing an underlying set-theoretic structure of points? Numerical identity (point-evaluation) is the degenerate case where all contexts contract to a single point.

   Paper I does this for the Boolean case: probability is reconstructed from directed refinement of finite Boolean algebras, without assuming a sample space. The open question is whether this extends beyond the Boolean case, and if so what plays the role of "directed refinement" when the algebra is not distributive.

   (Sharpened by Zafiris, Ch. 6 of Foundations of Relational Realism: the relational description is not a technical convenience but the starting point; point-spaces are what you reconstruct, not what you assume.)

3. **Local-to-global extension.**

   When does locally defined information (on each Boolean context) determine a global object? What are the *conditions on the site* — not the individual valuations — that control whether local data extend globally?

   CE is one instance: local charges always glue finitely additively (Kolmogorov), but gluing to a σ-additive measure requires something extra — and that "something extra" is not a sheaf condition on any known topology (Zafiris 2006, Biesel 2024: dictionary translation, not theorem). Paper II's EA/PR/VDR is another instance: the obstruction to global state extension in the non-Boolean case is non-distributivity rather than failure of σ-additivity.

   The general form: both the Boolean and non-Boolean extension problems are local-to-global questions on different categories. What controls the answer is the structure of the site, not the properties of individual states.

   (Sharpened by Zafiris, Ch. 6: the gluing conditions on overlapping Boolean reference frames are the structural content; the topology of the site is what does the work.)

Part I
The extension boundary

The two surviving open problems of the programme: Strategy D (Boolean, set-theoretic topology) and the OML extension problem (non-Boolean, algebraic). Each asks what structural conditions force honest (σ-additive) probability.

1 Strategy D

Synthesis: the σ-complete characterization pattern

Every known *characterization theorem* for when a Boolean algebra carries a strictly positive σ-additive probability presupposes Dedekind σ-completeness:

- Kelley (1959), Theorem 9: complete + weakly distributive + chargeable.
- KVP (Fremlin 391D): Dedekind σ-complete + weakly (σ, ∞)-distributive + chargeable.
- Balcar–Jech–Pazák (Fremlin 539N, under p-ideal dichotomy): Dedekind complete + ccc $\Rightarrow$ Maharam algebra.
- Kalton–Roberts (Fremlin 392F–G): uniformly exhaustive submeasure + Dedekind σ-complete.
- All negative constructions (Argyros 1983, Souslin line under $\neg$SH, Farah–Veličković 2006, Talagrand 2008) also operate in the σ-complete row.

The non-σ-complete regime has no analogous characterization — only piecewise existence results for specific construction families (minimally generated algebras, Borodulin-Nadzieja; Theorem 9.4 in Plebanek 2024). This makes Strategy D structurally novel: not "unstudied," but *no measure-existence characterization reaches here.* Confirmed across ≥ 6 sources: Fremlin ch39, ch53, ch54; Kelley (1959); Argyros (1983); Džamonja–Plebanek (2008).

The most concrete ZFC construction is Todorčević's example (Džamonja–Plebanek, Theorem 3.1): a ccc, non-σ-centred, measure-free Boolean algebra of cardinality $\mathrm{add}(\mathcal{N})$. However, the generating family $\{T_a\}$ forms a chain under $\subseteq^*$, making the generated algebra an interval algebra with atoms (see the detailed entry below). $\mathfrak{A}$ is **not** a Strategy D counterexample. It demonstrates that measure-freeness does not require σ-completeness, but no ZFC atomless non-σ-complete measure-free example is known.

Surveys and orientation

Plebanek — "A survey on topological properties of $P(K)$ spaces" (2024)

READ (2026-05-20). Verdict: most current secondary source for Strategy D. Confirms the exact parameter regime is open. Identifies near-miss constructions and a potential construction family (minimally generated Boolean algebras).

Source: arXiv:2312.14755v3 (8 Jul 2024), 33 pages.

Scope. Surveys topological properties of $P(K)$, the space of Radon probability measures on a compact Hausdorff space K, equipped with the weak* topology inherited from $C(K)^*$. For metrizable K, the structure of $P(K)$ is well understood ($P(K) \cong [0,1]^\omega$ for infinite K). For nonmetrizable compacta, even basic properties depend on additional axioms of set theory. The survey focuses on this nonmetrizable regime.

Strategy D assessment. Strategy D asks: does there exist a compact totally disconnected space K such that (i) K has no isolated points (= non-atomic Boolean algebra), (ii) K is not basically disconnected (= non-σ-complete Boolean algebra), and (iii) K carries no strictly positive Radon probability?

The survey does not pose this exact question but provides the landscape in which it sits.

Condition table for candidate constructions. Each candidate is checked against Strategy D's three conditions: non-atomic (✓/×/?), non-σ-complete (✓/×/?), measure-free (✓/×/?).

Construction	**Non-at.**	**Non-σ-c.**	**Meas-free**	**Note**
Argyros (1983, CH)	✓	×	✓	basically disconnected
Kunen (1981, CH)	✓	✓	×	compact L-space; carries s.p. normal measur
Fedorchuk (1975, $\diamondsuit$)	✓	✓	?	hereditarily separable; measure status unclea
Talagrand (1980)	✓	✓	?	Corson compact; has s.p. measure (Cor. 5.3)
Minimally generated	✓	✓	?	Borodulin-Nadzieja; see below

No candidate satisfies all three conditions. Argyros's construction is the closest to measure-free but fails condition (ii): it is basically disconnected. Kunen's compact L-space satisfies (i) and (ii) but the survey does not determine (iii); this is the most promising near-miss to investigate.

Minimally generated Boolean algebras (Theorem 9.4, citing Borodulin-Nadzieja). A Boolean algebra A is *minimally generated* if it is the closure of a chain of simple (= one-generator) extensions starting from the two-element algebra. Such algebras are typically non-σ-complete and non-atomic. Their Stone spaces have controlled topological structure amenable to combinatorial analysis. Theorem 9.4 characterizes when a minimally generated BA admits a strictly positive measure of *countable Maharam type* (via an increasing sequence of regular measures on the generating chain).

Why this matters for Strategy D. Minimally generated algebras automatically satisfy conditions (i) and (ii). If one could construct a minimally generated algebra failing the hypothesis of Theorem 9.4, it would be a Strategy D candidate — provided the failure of countable-type s.p. measures extends to s.p. Radon measures of all types. Caveat: Theorem 9.4 characterizes measures of *countable type* specifically; absence of a countable-type s.p. measure is weaker than absence of *any* s.p. Radon measure.

Baire vs. Radon distinction. The survey works throughout with *Radon* measures on K: σ-additive on the Borel σ-algebra $\mathrm{Bor}(K)$, inner regular with respect to compact sets. Paper I's stone_measure_exists extends a finitely additive charge on a Boolean algebra C to a Baire measure on $\mathrm{St}(C)$ (the σ-algebra generated by clopens). For metrizable Stone spaces, Baire = Borel and every finite Baire measure is Radon, so the distinction collapses. For nonmetrizable compacta (all Strategy D candidates), Baire $\subsetneq$ Borel and a Baire measure need not extend uniquely to a Radon measure.

This distinction matters: Strategy D asks about strictly positive *Radon* measures. Paper I produces *Baire* measures. The step from Baire to Radon is automatic in metrizable spaces but requires additional argument in the nonmetrizable regime where Strategy D lives.

Key results for Strategy D orientation:

- **Cor. 2.8:** K carries a strictly positive Radon measure iff there exists a continuous surjection from K onto a metrizable space with a strictly positive measure. This is the Radon-level analogue of the Kelley intersection-number condition.
- **Thm. 4.5:** under MA+¬CH, every compact space of weight $< \mathfrak{c}$ carrying a strictly positive measure of countable Maharam type is a continuous image of the Čech–Stone remainder $\omega^* = \beta\omega \setminus \omega$. This gives structural constraints on the topology of spaces admitting such measures.
- **§3.2 (Haydon's problem):** "Given $\mu \in P(K)$ and an uncountable family of Borel sets with positive measure, is there a point in K belonging to uncountably many of those sets?" Consistently yes (under PFA), consistently no (under CH). Set-theoretic sensitivity of the entire subject.
- **§7:** recent results on measure-theoretic structure of Stone spaces of Boolean algebras, including conditions for $P(K)$ to be a Valdivia or Corson compact.

What the survey does NOT provide:

- No construction satisfying all three Strategy D conditions.
- No proof that such a construction is impossible.
- No discussion of the non-σ-complete / non-basically-disconnected regime as a separate question.
- No consistency result (positive or negative) for Strategy D's exact parameter regime.

Contact with programme. Confirms Strategy D is open and positions it precisely: the question sits in the gap between Argyros's basically-disconnected measure-free constructions and the general theory of s.p. Radon measures on zero-dimensional compacta. The minimally generated BA lineage (Borodulin-Nadzieja, Theorem 9.4) is the most promising construction family to investigate. Kunen's compact L-space is the most promising near-miss to check against condition (iii).

Next steps suggested:
1. Read Plebanek (2023), "Musing on Kunen's compact L-space" (arXiv:2012.01849, 10pp) to determine whether Kunen's construction satisfies condition (iii).
2. Read Argyros (1983) to verify exactly which conditions it satisfies and confirm that basic disconnectedness is essential to the construction.
3. Read Džamonja–Plebanek (2008) for chain-condition characterizations of s.p. measure existence.

Plebanek — "Musing on Kunen's compact L-space" (2023)

READ (2026-05-20). Verdict: definitively settles Kunen's L-space as $\times$ on Strategy D condition (iii). Structural lesson: the entire L-space approach to Strategy D is consistency-only, never ZFC.

Source: arXiv:2012.01849 (published *Topology and its Applications* 330, 2023), 10 pages.

The construction. Under $\mathrm{cf}(\mathfrak{c}) = \omega_1$, Plebanek constructs a compact connected Corson compact L-space K carrying a strictly positive normal probability measure μ. "Normal" means $\mu(B) = 0$ for every Borel set B with empty interior (Plebanek's Definition 1.1); since $K = \mathrm{supp}(\mu)$, the measure is strictly positive. The space is an L-space: hereditarily Lindelöf but not separable.

Strategy D verdict. Kunen's compact L-space satisfies conditions (i) (no isolated points, since connected) and (ii) (not basically disconnected — in fact not even zero-dimensional, since connected). But condition (iii) *fails decisively*: the construction explicitly produces a strictly positive Radon probability by design. This is not a gap to be investigated; it is the construction's central purpose.

The Juhász constraint. Under MA+¬CH, no compact L-spaces exist (Juhász, classical). This means the entire Kunen/L-space lineage of constructions is *consistency-only*: even if one could modify a Kunen-type construction to remove the strictly positive measure (which would mean removing its raison d'être), the resulting space would not exist in ZFC. For Strategy D, this rules out attack mode A (ZFC construction) via the L-space route entirely. Any ZFC Strategy D candidate must come from a different construction family.

Additional structural results:
- **Thm. 3.4:** K has no nonmetrizable zero-dimensional closed subspaces. This makes K structurally remote from the totally disconnected setting where Strategy D lives.
- **Thm. 4.3:** $C(K)$ is not isomorphic to $C(L)$ for any zero-dimensional L. The Banach-space structure of Kunen's space is incompatible with zero-dimensional compacta.

Contact with programme. Closes the Kunen near-miss: $\times$ on condition (iii), with the stronger lesson that the L-space approach is consistency-only by the Juhász constraint. The condition table in the Plebanek (2024) entry is updated accordingly. The remaining near-misses to investigate are Fedorchuk ($\diamondsuit$, measure status unclear), Talagrand (Corson compact, likely has s.p. measure), and minimally generated BAs (Borodulin-Nadzieja, the most promising family).

Constructions and examples

Kelley — "Measures on Boolean algebras" (1959)

Why read this. Foundational paper on the existence of measures on Boolean algebras. Introduces the intersection number and the Kelley criterion (Theorem 4), the key chargeability characterisation. Also the first source for the link between σ-additivity and weak distributivity in complete algebras.

What to read. All 14 pages; the paper is short and self-contained.

Key results for Strategy D.

- **Theorem 4 (Kelley criterion).** A Boolean algebra B carries a strictly positive finitely additive measure iff $B \setminus \{0\}$ is a countable union of classes each with positive intersection number. This is chargeability = Fremlin 391J. Chargeability is necessary but not sufficient for σ-additive measures; Strategy D asks about the σ-additive level.
- **Theorem 9.** A *complete* Boolean algebra carrying a strictly positive measure admits a strictly positive σ-additive measure iff it is weakly (ω_1, ω)-distributive. This is the KVP theorem for the complete case. The completeness hypothesis is load-bearing and cannot be dropped.
- **Theorem 12 (Souslin line).** Assuming the negation of the Souslin hypothesis, the regular-open algebra of a Souslin line is complete, atomless, ccc, weakly countably distributive, and measure-free. Against Strategy D conditions: satisfies (i) non-atomic and (iii) measure-free, but fails (ii) because it is complete (hence σ-complete). Same row as Argyros.
- **Ryll-Nardzewski addendum** (end of paper). Gives a σ-measure analogue of Theorem 4: a monotone σ-approximation condition. Less well-known than the Kelley criterion; potentially relevant if one wants to characterise σ-additive measures directly rather than passing through chargeability + weak distributivity.

What it does *not* tell you. Every σ-additivity result (Theorems 8, 9, 12) assumes completeness. Kelley does not address the non-σ-complete regime at all. The paper establishes the framework (intersection number, chargeability, weak distributivity) but leaves Strategy D's exact parameter regime untouched.

Questions to carry forward.

1. Does the Ryll-Nardzewski σ-approximation condition have content in the non-σ-complete case, or does it degenerate?
2. Is there a meaningful weakening of weak distributivity that applies to non-σ-complete algebras?
3. The Souslin-line example (Theorem 12) is consistency-only ($\neg$SH). Are there ZFC complete measure-free examples? (Argyros gives one under CH; Talagrand's Maharam algebra is ZFC but asks a different question.)

Argyros — "On compact spaces without strictly positive measure" (1983)

Why read this. Constructs compact Hausdorff spaces with property $(*)$ (a chain condition stronger than ccc) but no strictly positive measure. These are the nearest known constructions to a Strategy D counterexample.

What to read. All 19 pages, focusing on the main construction (the Stone–Čech compactification βY route) and the verification of property $(*)$.

Key results for Strategy D.

- **Main construction.** Builds compact spaces $X_n = \beta Y_n$ (Stone–Čech compactification of a locally compact space Y_n) satisfying property $(*)$ but carrying no strictly positive Radon measure. This gives condition (iii) — measure-free.
- **Condition (i).** The spaces X_n have no isolated points (verified in the construction), so the corresponding clopen algebras are atomless. Satisfies condition (i).
- **Condition (ii) — fails.** The clopen algebra of βY_n is basically disconnected, i.e., σ-complete as a Boolean algebra. Plebanek (2024 survey, Table 1) lists Argyros's spaces as basically disconnected. Thus Argyros satisfies (i) + (iii) but fails (ii). Same row as Kelley's Souslin-line example: the wrong row of the hierarchy.
- **Set-theoretic dependence.** The construction appears to use CH or weaker fragments. The status under MA + $\neg$CH is unclear from this paper alone.

Methodological warning. The Stone–Čech compactification route naturally produces basically disconnected (or even extremally disconnected) spaces, because βY inherits strong completeness properties. This is a systematic obstruction: the standard technique for building measure-free spaces also forces σ-completeness. Any Strategy D construction must avoid this route or find a way to break the basic-disconnectedness that βY provides.

Questions to carry forward.

1. Is there an Argyros-style construction that avoids Stone–Čech compactification and produces a non-basically-disconnected space?
2. Can one take a quotient or subalgebra of Argyros's algebra that breaks σ-completeness while preserving measure-freeness?
3. The pattern: every known measure-free construction (Argyros, Souslin line, Gaifman) is σ-complete. Is this a coincidence or a structural obstruction?

Džamonja–Plebanek — "Strictly positive measures on Boolean algebras" (2008)

READ (2026-05-20). Verdict: most directly relevant paper for Strategy D. Contains the Todorčević ZFC example (Theorem 3.1) whose Strategy D status reduces to two verifiable conditions on an explicit construction.

Source: *Journal of Symbolic Logic* 73(4), 1416–1432.

Key results for Strategy D.

- **§1 (ZFC counterexample to approximability).** Starting from a Simon space (UDSP), constructs a Boolean algebra $\mathfrak{B}$ that is approximable (Claim 1.4) but admits no separable strictly positive measure (Claim 1.5). This refutes the conjecture that Talagrand's "approximability" condition suffices for separable s.p. measures. The algebra *does* carry a (non-separable) strictly positive measure, so this is not measure-free.
- **§2, Theorem 2.1 (ccc forcing).** For every σ-finite cc Boolean algebra $\mathfrak{B}$, there is a ccc forcing making it approximable (and of size $\leq \mathfrak{c}$). If $\mathfrak{B}$ is atomless, each measure in the approximating sequence is nonatomic.
- **Corollary 2.8 (MA + $\neg$CH).** Every atomless ccc Boolean algebra of size $< \mathfrak{c}$ carries a strictly positive nonatomic measure. Consistency result *against* Strategy D in the small-algebra regime.
- **Theorem 2.9 (chain condition characterisation).** $\mathfrak{A}$ carries a strictly positive nonatomic measure iff $\mathfrak{A}^+ = \bigcup_n B_n$ where: (i) $B_n \subseteq B_{n+1}$, (ii) $\mathrm{int}(B_n) \geq 2^{-n}$, (iii) every $b \in B_n$ has disjoint $b_1, b_2 \in B_{n+1}$ below b.
- **Corollary 2.11 (MA + $\neg$CH).** For atomless Boolean algebras of size $< \mathfrak{c}$, ccc $\iff$ the chain condition of Theorem 2.9.
- **Theorem 3.1 (Todorčević example, ZFC).** There exists (in ZFC) a Boolean algebra $\mathfrak{A}$ of size $\mathrm{add}(\mathcal{N})$ that is ccc, not σ-centred, and carries no strictly positive measure.

Satisfies Strategy D condition (iii) outright.

- **Theorem 3.3 (consistency, $\mathrm{cof}(\mathcal{N}_\nu) = \omega_1$).** There is a Boolean algebra of cardinality ω_1 with a strictly positive measure but no *separable* strictly positive measure.

The Todorčević construction (Theorem 3.1, traced to source).

Džamonja–Plebanek extract Theorem 3.1 from Todorčević's Theorem 8.4 in "Chain-condition methods in topology," *Topology and Its Applications* 101 (2000), 45–82. The construction (now read in full):

Let K be the Cantor set, $Z = \{x \in K : \lim |x[i]|/i = 0\}$ (zero-density elements). By a Ramsey argument, there exist $A \subseteq Z$ (well-ordered under $\subseteq^*$, order-type $= \mathrm{add}(\mathcal{N})$) and B (a tower with uncountable coinitiality). Set $T = \{(t, n) : n \in \mathbb{N},\ t \subseteq n\}$, and define generators:

$$T_a = \{(s, m) \in T : a \cap m \subseteq s\}, \qquad T_{(t,n)} = \{(s, m) \in T : m \geq n,\ s \cap n = t\}.$$

Let $\mathcal{B}$ be the subalgebra of $\mathcal{P}(T)/\mathrm{fin}$ generated by both kinds. Todorčević proves: $\mathcal{B}$ is ccc (Knaster), not σ-centred, every ultrafilter has countable π-character, and every quotient $\mathcal{B}/\mathcal{U}$ is an interval algebra of order-type $< \mathrm{add}(\mathcal{N})$.

Džamonja–Plebanek take the subalgebra $\mathfrak{A}$ generated by the T_a's **alone** ("generators of the first kind"). They verify: $\mathfrak{A}$ is ccc and not σ-centred (inherited), and Lemma 3.2 shows no strictly positive measure exists (the generating family $\{T_a\}$ satisfies $a \leq b \Rightarrow T_a \cdot T_b = 0$ in the quotient, forcing any measure to vanish on a dense set).

Strategy D verification status (2026-05-20). RESOLVED: $\mathfrak{A}$ has atoms.

The generating family $\{[T_a] : a \in A\}$ forms a *chain* in $\mathcal{P}(T)/\mathrm{fin}$: since A is totally ordered under $\subseteq^*$ and $a \subseteq^* b$ implies $T_b \subseteq^* T_a$, the map $a \mapsto [T_a]$ reverses the $\subseteq^*$-order. A Boolean algebra generated by a single chain is an interval algebra. Every element of $\mathfrak{A}$ is a finite union of "intervals" $[T_a] \setminus [T_{a'}]$ where $a \subseteq^* a'$ in A.

Since A has order-type $\mathrm{add}(\mathcal{N})$ (an ordinal), every non-maximal $a \in A$ has an immediate $\subseteq^*$-successor a'. Then $[T_a] \setminus [T_{a'}]$ is:

- *Nonzero:* $a' \setminus a$ is infinite (since $a \subsetneq^* a'$), so infinitely many $(s, m) \in T$ satisfy $a \cap m \subseteq s$ but $a' \cap m \not\subseteq s$.
- *An atom:* any sub-interval $[T_p] \setminus [T_q] \leq [T_a] \setminus [T_{a'}]$ requires $a \subseteq^* p \subseteq^* q \subseteq^* a'$. Immediate succession forces $p = a$, $q = a'$.

$\mathfrak{A}$ fails Strategy D condition (i): it is not atomless. Džamonja–Plebanek's Theorem 3.1 does not claim atomlessness — they state $|\mathfrak{A}|$, ccc, not σ-centred, and the chain/disjointness generator condition, but atomlessness is not among the conclusions.

What survives. $\mathfrak{A}$ is the first known ZFC example of a ccc Boolean algebra with no strictly positive measure and cardinality $\mathrm{add}(\mathcal{N})$. It demonstrates that measure-freeness can arise in $\mathcal{P}(\omega)/\mathrm{fin}$ subalgebras (outside the β-compactification route that forces σ-completeness). But it is *not* a Strategy D counterexample.

Remaining candidates. Todorčević's full algebra $\mathcal{B}$ (with both T_a and $T_{(t,n)}$ generators) might be atomless — the $T_{(t,n)}$'s have a tree structure and Claim 1 of Theorem 8.4 shows every positive element contains some $T_a \cap T_{(t,n)}$, consistent with splitting. But $\mathcal{B}$ lacks the no-s.p.-measure proof: Lemma 3.2 used the chain property of the T_a-only subalgebra. Whether $\mathcal{B}$ carries a strictly positive measure is a separate question.

Synthesis observations.

- The MA + ¬CH results (Corollaries 2.8 and 2.11) show that under MA + ¬CH, every atomless ccc algebra of size $< \mathfrak{c}$ carries a strictly positive nonatomic measure. This pushes Strategy D toward set-theoretic independence.
- Every known ZFC measure-free construction (Argyros, Souslin line, Gaifman, and now $\mathfrak{A}$) either is σ-complete or has atoms. No ZFC atomless non-σ-complete measure-free example is known.

- The structural reason: the standard technique (β-compactification) forces σ-completeness; the $\mathcal{P}(\omega)$/fin subalgebra route (Todorčević) avoids σ-completeness but the chain-generated subalgebra has atoms. Obtaining both atomlessness and measure-freeness simultaneously in ZFC remains open.

Remaining questions.

1. Does the pattern "ZFC measure-free $\Rightarrow$ σ-complete or atomic" admit a structural explanation, or is it an artifact of the available constructions?
2. Can the MA + $\neg$CH positive results (Corollary 2.8) be strengthened to cover all atomless ccc algebras, not just those of size $< \mathfrak{c}$?
3. Does Todorčević's full $\mathcal{B}$ carry a strictly positive measure? If not, is it atomless?

Fremlin reference chapters

Fremlin Vol. 5, Ch. 53 — topologies and measures III

Source: Fremlin, *Measure Theory* Vol. 5, Ch. 53, §§ 531–539 (2008).

Why read this. Strategy D asks whether a non-σ-complete non-atomic measure-free Boolean algebra exists. Chapter 53 is the definitive modern treatment of the interaction between topological properties and measure existence on compact spaces and Boolean algebras. The question is whether any of its powerful results reach the non-σ-complete regime.

What the chapter covers.

- §531: Maharam types and Maharam-algebra structure theory. Operating assumption: Dedekind complete.
- §532: completion regular measures. Focus on extending measures from subalgebras to completions — not directly relevant, but confirms the machinery assumes σ-completeness throughout.
- §533: measure-compactness and measure-compact cardinals. Key concept: a compact space K is *measure-compact* if every Baire measure extends to a Radon measure. Results characterize when compact spaces carry measures with prescribed support, but the Boolean algebras involved are σ-complete or complete.
- §534: Hausdorff measures and strong measure zero. Metric setting; not relevant to Strategy D's algebraic question.
- §535–538: Liftings, density topologies, pointwise compact sets of measurable functions. Background infrastructure.
- §539: **Key section.** The Balcar–Jech–Pazák theorem and surrounding results on the relationship between weak (σ, ∞)-distributivity and Maharam-algebra structure.

§539 in detail. The main results:

- 539A–539C: A Dedekind complete Boolean algebra is a Maharam algebra iff it is weakly (σ, ∞)-distributive and carries a strictly positive Maharam submeasure.
- 539N (Balcar–Jech–Pazák under p-ideal dichotomy): every weakly (σ, ∞)-distributive Dedekind complete ccc Boolean algebra is a Maharam algebra.
- 539Qg (Farah–Veličković 2006): under $2^\kappa = \kappa^+$ and $\square_\kappa$, there exist Dedekind complete ccc algebras that are NOT Maharam algebras (i.e., non-measurable despite Dedekind completeness + ccc).

Verdict for Strategy D: *Every* result in §539 assumes Dedekind σ-completeness or full Dedekind completeness. No theorem, example, or remark addresses Boolean algebras that are non-σ-complete. This is now confirmed across ≥ 5 sources: Fremlin ch39, Fremlin ch53, Kelley (1959), Argyros (1983), Džamonja–Plebanek (2008) — all measure-theoretic results on Boolean algebras operate in the σ-complete row. Strategy D's non-σ-complete regime is genuinely uncharted.

Questions.

1. Does weak (σ, ∞)-distributivity have a meaningful analogue for non-σ-complete Boolean algebras?
2. The Farah–Veličković consistency result shows Dedekind-complete ccc algebras can be non-measurable. Could a variant construction drop the completeness requirement?
3. Can the Balcar–Jech–Pazák machinery be adapted to "partial σ-completeness" (joins exist for some but not all countable families)?

Fremlin Vol. 5, Ch. 54 — real-valued-measurable cardinals

Source: Fremlin, *Measure Theory* Vol. 5, Ch. 54, §§ 541–544 (2008).

Why read this. Strategy D asks whether a non-σ-complete non-atomic measure-free Boolean algebra exists. Chapter 54 is the definitive treatment of when measures can live on power sets and their quotients. The question is whether any of its results constrain the Strategy D parameter regime.

What the chapter covers.

- §541: Saturated ideals. An ideal $\mathcal{I}$ of $\mathcal{P}\kappa$ is κ-saturated and κ-additive. Key result: quotient $\mathcal{P}\kappa/\mathcal{I}$ is always Dedekind complete (541B). Ulam's dichotomy (541P): the quotient algebra is either atomless with $\kappa \leq \mathfrak{c}$ (atomlessly-measurable), or purely atomic with $\kappa > \mathfrak{c}$ (two-valued-measurable).
- §542: Quasi-measurable cardinals. Cardinal arithmetic under measurability hypotheses.
- §543: The Gitik–Shelah theorem. If $(X, \mathcal{P}X, \mu)$ is a probability space with singletons negligible, the Maharam type of μ must be large ($\geq$ the additivity of μ). Key concept: a cardinal λ is *measure-free* iff there is no real-valued-measurable $\kappa \leq \lambda$.
- §544: Atomlessly-measurable cardinals. Repeated integration, null ideals, witnessing probabilities.

Verdict for Strategy D: *Not directly relevant.* The entire chapter operates on power-set algebras $\mathcal{P}X$ and their quotients $\mathcal{P}X/\mathcal{I}$, which are always Dedekind complete (541B). The "measure-free" concept (543B(e)) concerns cardinals, not general Boolean algebras. Strategy D asks about specific non-σ-complete Boolean algebras, which are not power-set quotients.

What it confirms. The chapter reinforces the pattern from ch53: the most powerful measure-existence and measure-nonexistence results in Fremlin operate in the Dedekind complete/σ-complete regime. The non-σ-complete world that Strategy D inhabits has no representation in this chapter.

One indirect connection. If a Boolean algebra B is non-σ-complete but can be embedded into a Dedekind complete algebra $\overline{B}$ preserving some structural property, then ch54 results about $\overline{B}$ might constrain what happens on B. But the embedding must preserve the relevant measure-theoretic properties, and it is unclear whether such embeddings exist in the Strategy D regime.

Fremlin Vol. 3, Ch. 39 — measurable algebras

READ (2026-05-17 and 2026-05-20). Verdict: answers the "what forces σ-additivity" question precisely for the σ-complete case. Weak (σ, ∞)-distributivity is the algebraic CE condition. The submeasure/control-measure extension (§§392–394) confirms the machinery stays in the σ-complete row.

Source: Fremlin, *Measure Theory* Vol. 3, Ch. 39, freely available at `https://www1.essex.ac.uk/maths/peopl`

391D Theorem (Kantorovich–Vulikh–Pinsker 1950). A Boolean algebra $\mathfrak{A}$ is measurable iff:

1. $\mathfrak{A}$ is Dedekind σ-complete,
2. $\mathfrak{A}$ is weakly (σ, ∞)-distributive,
3. $\mathfrak{A}$ is chargeable (admits a strictly positive finitely additive functional).

391J Theorem (Kelley). $\mathfrak{A}$ is chargeable iff $\mathfrak{A} = \{0\}$ or $\mathfrak{A} \setminus \{0\}$ is a countable union of sets with non-zero intersection numbers.

Contact with programme. This is the algebraic answer to Question 3 (local-to-global extension) for the Boolean case. The condition controlling whether local charges extend to a σ-additive global measure is a property of the *algebra* (the site), not the charges (the sections) — exactly what the programme's framing predicts. Weak (σ, ∞)-distributivity is the algebraic face of CE.

The submeasure extension (§§392–394). The KVP theorem characterizes measurable Boolean algebras. §§392–394 probe whether submeasure conditions can replace or supplement the KVP conditions — the "control measure problem." This entire investigation presupposes Dedekind σ-completeness.

Key results:

- **392F (Kalton–Roberts).** Uniformly exhaustive strictly positive submeasure ⇒ chargeable.
- **392G.** Measurable iff Dedekind σ-complete + weakly (σ, ∞)-distributive + admits a strictly positive uniformly exhaustive submeasure.
- **393 (Maharam submeasures).** A *Maharam algebra* = Dedekind σ-complete BA with a strictly positive continuous-at-0 submeasure. Every measurable algebra is Maharam.
- **394 (Talagrand 2008).** There exist Maharam algebras that are not measurable. The control measure problem is resolved negatively.

Relevance to Strategy D. The submeasure/control-measure landscape is the σ-complete neighbour of Strategy D, not Strategy D itself. All results assume Dedekind σ-completeness. Strategy D operates in the non-σ-complete regime. The conceptual lesson is that measurability conditions come in layers (chargeability < Maharam submeasure < uniformly exhaustive < measurability), and these layers separate even in the σ-complete case.

2 OML extension problem

Synthesis: the OML obstruction landscape

The OML extension problem asks: what forces σ-additive measure extension on a non-Boolean OML? The primary sources reveal a coherent obstruction landscape:

1. **The obstruction bites at the FA level** (Pták–Pulmannová 1994): subadditive unitality forces Boolean. Any condition strong enough to give full measure-like behavior destroys non-Boolean structure.
2. **The worst case exists** (Pták 1987): exotic OMLs have no states at all; states cannot extend from Boolean sublogics. The σ-complete case is partially resolved but open in general.
3. **Positive results require von Neumann–type completeness**: Gleason ($L(\mathcal{H})$, dim $\geq$ 3), Bunce–Wright (vN algebras, no Type I_2), Chetcuti–Dvurečenskij (lattice effect algebras). All stay within vN / JBW / effect-algebra completeness.
4. **Intermediate conditions fail**: regularity does not force σ-additivity on general OMPs (Navara 1992).
5. **The question is now well-posed** (McDonald–Bimbó 2023): Stone-type duality for OMLs provides the target space $\mathcal{S}_0(\mathfrak{A})$ on which to ask whether states extend to σ-additive measures. Before 2023, the OML extension question lacked a precise duality-theoretic formulation.

The gap between "too weak" (regularity, non-unital states) and "too strong" (subadditive unitality ⇒ Boolean; vN completeness ⇒ positive) appears structurally robust. Any new positive result must exploit special structure that exotic logics lack — Hilbert-space geometry, operator-algebraic completeness, or a genuinely new technique (categorical, topos-theoretic, or novel pasting).

Obstructions

Pták–Pulmannová (1994) — subadditive measures and orthomodularity

Source: P. Pták and S. Pulmannová, "A measure theoretic characterization of Boolean algebras among orthomodular lattices," *Czechoslovak Mathematical Journal* 44(119), 1994, pp. 743–746.

Why read this. The OML extension problem asks what structural conditions force honest (σ-additive) probability on a non-Boolean OML. This paper identifies the precise obstruction — and the obstruction bites already at the finitely additive level, before σ-additivity enters the picture.

Main result (Theorem 1). An orthomodular lattice L is Boolean if and only if L is *unital with respect to subadditive states*: for every $a \neq 0$ in L, there exists a state s on L with $s(a) = 1$ such that s is subadditive (i.e., $s(a \vee b) \leq s(a) + s(b)$ for all a, b).

Proof chain.

1. Prop 2: a subadditive state on an OML is a *valuation* ($s(a \vee b) + s(a \wedge b) = s(a) + s(b)$ for all a, b).
2. Prop 5: a valuation s with $s(a) = 1$ forces a to be *central* (i.e., compatible with every element).
3. Theorem 1: unital w.r.t. subadditive states $\Rightarrow$ every element is central $\Rightarrow$ Boolean.

Why this kills naïve OML extension. The chain shows that *any* condition strong enough to give full "measure-like" behavior (subadditivity + unitality) forces the lattice to be Boolean. This is not a σ-additivity issue — the obstruction is already at the finitely additive level. A subadditive state on a non-Boolean OML cannot be unital, and therefore cannot fully probe the lattice's structure.

Prop 6 (valuationless OMLs). There exist OMPs (orthomodular posets) carrying no non-trivial valuation at all — not even a non-unital one. So the gap between "some states exist" and "enough states to probe structure" is not merely quantitative; some structures are valuationally empty.

Consequence for the OML extension problem. Any approach to forcing σ-additivity on a non-Boolean OML must either:

1. avoid requiring subadditivity (work with genuinely non-subadditive measures);
2. accept non-unitality (some elements are invisible to the measure theory);
3. bypass the state space entirely (topos-theoretic or categorical methods); or
4. exploit special algebraic structure (as Gleason does for $L(\mathcal{H})$, dim ≥ 3).

Questions.

1. Where exactly does the proof of Prop 5 use orthomodularity versus distributivity? Could a weaker lattice condition suffice?
2. Is the unital-subadditive condition equivalent to any known condition in the Greechie/pasting literature?
3. Does Gleason's theorem for $L(\mathcal{H})$ evade this obstruction because $L(\mathcal{H})$ fails to be unital w.r.t. subadditive states, or because the Hilbert-space states satisfy a different structural condition?

Pták (1987) — exotic logics

Source: P. Pták, "Exotic logics," *Colloquium Mathematicum* 54(1), 1987, pp. 1–7.

Why read this. An OML ("logic") is *exotic* if it has no states at all, and *nearly exotic* if it has exactly one state. This paper constructs both and proves universal embedding theorems: any logic embeds into an exotic one (Theorem 3.1), and any logic with a two-valued state embeds into a nearly exotic one (Theorem 3.2), each with prescribed Boolean center. The Section 4 appendix extends these to σ-complete logics.

READ (2026-05-17). Verdict: populates Square A bottom-right. Obstruction is combinatorial (Greechie pasting), not from the center.

The construction. An *exotic logic* is a σ-orthocomplete OMP that admits finitely additive states but no σ-additive states. Pták constructs these by *Greechie pasting*: start with a collection of Boolean algebras (blocks), identify certain atoms across blocks so that the resulting structure is an OMP, and show that any σ-additive state would require the atoms of one block to sum to more than 1 (by the identifications forcing double-counting).

Main results.

- **Greechie's exotic logic:** a finite OML with no states. Constructed from 20 three-atom and 3 four-atom Boolean blocks pasted so that the atom set admits two partitions into maximal Boolean subalgebras with 12 and 11 classes respectively — contradicting state existence.
- **Prop 2.1:** if L is exotic, then $L \times \{0, 1\}$ is nearly exotic (exactly one state).
- **Prop 3.1:** any logic embeds into an exotic logic with trivial center.
- **Theorem 3.1:** for any logic L_1 and Boolean algebra B, there exists an exotic logic L with L_1 as sublogic and $C(L) = B$.
- **Theorem 3.2:** if L_1 has a two-valued state, the same holds for nearly exotic L.
- **Corollary:** any Boolean algebra embeds into a nearly exotic logic with arbitrary center — ruling out state-extension from Boolean sublogics to the whole logic.

Section 4: σ-complete case. Theorem 4.1:

1. If L_1 is σ-complete and B is a σ-algebra of all subsets of a set, then L_1 σ-completely embeds into an exotic logic L with $C(L) = B$. If L_1 also has a two-valued σ-additive state, the same for nearly exotic.
2. If L_1 is σ-complete with all Boolean sublogics finite, same conclusion for arbitrary σ-Boolean B.

The general σ-complete case remains open — the construction does not always produce σ-complete logics. Pták notes this may be related to the existence of least σ-products (open problem in Sikorski).

Relevance to the OML extension problem. This paper establishes the *worst case* for OML state theory: state-free OMLs exist, states cannot be extended from Boolean sublogics, and the exotic/nearly-exotic phenomena persist in the σ-complete setting. Combined with Pták–Pulmannová (1994), the picture is:

- subadditive unitality forces Boolean (1994);
- without unitality, states can be entirely absent (1987);
- even one-state logics resist extension from Boolean parts.

Any positive OML extension result must exploit special structure (Hilbert-space geometry, von Neumann completeness) that exotic logics lack.

Questions.

1. Is the σ-complete exotic embedding problem (general case) still open, or has it been resolved since 1987?
2. What is the minimal size of an exotic OML? Greechie's original has ~ 70 atoms — is there something smaller?
3. Do the exotic embedding theorems extend to OMPs (orthomodular posets), or does the lattice requirement play a role?

Navara — "Regularity and σ-additivity of states" (1992)

READ (2026-05-17). Verdict: counterexample kills the conjecture that regularity forces σ-additivity. Not an admissibility condition.

Source: *Proc. AMS* 115(2), 427–429.

The result. Béaver and Cook (1989) proved: on *projection lattices of von Neumann algebras*, every P-regular state (= inner regular w.r.t. finite-dimensional projections) is σ-additive. Navara shows this does NOT generalize: there exist σ-orthocomplete quantum logics (OMPs) on which all states are P-regular yet some are not σ-additive.

The construction. Navara modifies Pták's exotic-logic construction: takes a Greechie pasting that admits only non-σ-additive states, then verifies that all states on this logic are P-regular (because the finite-dimensional sub-logics approximate the full logic sufficiently well for inner regularity, but not for σ-additivity).

Contact with programme: Clarifies the bottom-right cell of Square A. The OML extension problem does *not* have a simple admissibility condition analogous to regularity. The gap between finitely additive states and σ-additive states on OMPs is not bridgeable by regularity alone — the obstruction (Pták's combinatorial pasting) persists even when all states are regular. This confirms that the non-Boolean extension problem is structurally harder than the Boolean one (where weak (σ, ∞)-distributivity does the job via KVP 1950).

Positive results and state-space geometry

Shultz — "A characterization of state spaces of orthomodular lattices" (1974)

READ (2026-05-17). Verdict: orthomodularity imposes NO geometric constraint on state spaces; but σ-additive state spaces CAN be geometrically degenerate.

Source: *J. Combin. Theory A* 17(3), 317–328. DOI: 10.1016/0097-3165(74)90096-X.

Main Theorem (p. 321). If C is a compact convex subset of a locally convex space, then $C \cong S(L)$ (affinely homeomorphic to the state space) of some σ-orthomodular lattice L. If C is a rational polytope, L can be chosen finite.

Construction. Uses Greechie's G-spaces (the same pasting technique as Pták's exotic logics): points + frames with loop constraints. Lemma 1: any G-space yields a σ-OML whose state space is affinely homeomorphic to the weight space $W(X, \mathscr{F})$. Lemmas 2–3: forcing techniques (adjoining points, controlling weight values) to realize arbitrary compact convex sets as weight spaces.

The negative result for σ-states (p. 327). The converse is FALSE for σ-additive states. Take L = Borel subsets of $\mathbb{R}$ modulo Lebesgue-null sets (a σ-Boolean algebra). Then $S_\sigma(L)$ has no extreme points — by Krein–Milman, $S_\sigma(L)$ is NOT affinely equivalent to any compact convex subset of a locally convex space. The σ-state space can be geometrically degenerate in ways the full state space cannot.

Contact with programme: Square A, bottom row.

1. Orthomodularity alone imposes NO constraint on state-space geometry — any compact convex set arises. The obstruction to σ-additivity (Pták) cannot be read off from convex geometry.
2. $S_\sigma(L)$ is structurally different from $S(L)$: it can lack extreme points entirely. This is not just "a smaller convex set" — it can fail to be a compact convex set at all in the relevant sense.
3. Greechie pasting is the common tool behind both Shultz (state-space universality) and Pták (exotic logics with no σ-states). The pasting is the fundamental technique for the non-Boolean column.

Bunce–Wright (1992) — the Mackey–Gleason problem

Source: L. J. Bunce and J. D. M. Wright, "The Mackey–Gleason Problem," *Bulletin of the AMS* 26(2), 1992, pp. 288–293.

Why read this. The Mackey–Gleason problem asks: does every bounded finitely additive measure on the projections of a von Neumann algebra extend to a bounded linear functional on the algebra? This is the operator-algebraic version of the OML extension problem.

Main result (Theorem). Let $\mathcal{A}$ be a von Neumann algebra with no direct summand of Type I_2. Then every bounded finitely additive measure on $\mathrm{Proj}(\mathcal{A})$ extends uniquely to a bounded linear functional on $\mathcal{A}$.

Scope and limits.

- The extension target is a *bounded linear functional on the von Neumann algebra*, not a σ-additive measure on a dual space like $S_0(\mathcal{A})$. These are different questions.
- The von Neumann algebra structure is load-bearing: the proof uses the Jordan decomposition, spectral theory, and the Kadison–Singer paving machinery. None of this is available for a general OML.
- The Type I_2 exclusion is genuine: $M_2(\mathbb{C})$ has "exotic" measures on its projections (the 2×2 case is the Kochen–Specker boundary).
- For the specific case of σ-additive measures on projections (normal states), the result is Gleason's theorem (dim ≥ 3). Bunce–Wright generalizes from σ-additive to bounded FA, which is the harder direction.

What this does NOT do for the OML extension problem. The OML extension problem asks: given a state s on a general OML A, does s extend to a σ-additive measure on the McDonald–Bimbó dual space $S_0(A)$? Bunce–Wright addresses:

1. a different source structure (von Neumann projections, not general OML);
2. a different extension target (bounded linear functional, not σ-additive measure on dual space);
3. a different direction (FA $\to$ linear functional, not state $\to$ σ-additive measure).

What it confirms. Positive extension results require rich algebraic structure. The progression Gleason $\to$ Bunce–Wright $\to$ Chetcuti–Dvurečenskij (2003) tracks increasing generality, but all stay within the von Neumann / JBW / lattice-effect-algebra world where completeness and spectral theory are available.

Questions.

1. Is there a purely lattice-theoretic characterization of when Bunce–Wright-type extension works, without reference to the ambient C^*-algebra?
2. Does the Type I_2 exclusion correspond to a lattice property visible in the OML setting (e.g., height ≤ 2 blocks)?
3. Does the resolution of the Kadison–Singer conjecture (Marcus–Spielman–Srivastava 2015) have downstream consequences for measure extension on broader classes of OMLs, or is it confined to the paving problem?

Duality framework

McDonald–Bimbó (2023) — Stone-type duality for OMLs

Source: J. McDonald and K. Bimbó, "Stone-type duality for orthomodular lattices," manuscript, 2023.

Why read this. The classical Stone duality (1936) represents Boolean algebras as clopen algebras of Stone spaces. This paper extends the duality to orthomodular lattices, providing the topological dual space $S_0(\mathfrak{A})$ that makes the OML extension question well-posed: "does a state on $\mathfrak{A}$ extend to a σ-additive measure on $S_0(\mathfrak{A})$?"

Construction. For an OML $\mathfrak{A}$:

- $\mathcal{S}_0(\mathfrak{A})$ is the set of all *filters* of $\mathfrak{A}$ (not just ultrafilters, as in classical Stone duality).

- Topology: generated by sets $\sigma_0(a) = \{F \in \mathcal{S}_0(\mathfrak{A}) : a \in F\}$ for $a \in \mathfrak{A}$.
- $\mathcal{A}_0(\mathcal{S}_0(\mathfrak{A}))$: the algebra of clopen "orthogonality-respecting" subsets of the dual space.

Main results.

- Theorem 3.9: $\mathfrak{A} \cong \mathcal{A}_0(\mathcal{S}_0(\mathfrak{A}))$ — full algebraic recovery.
- Theorem 3.11: algebraic realization — every "OML-space" arises as $\mathcal{S}_0$ of some OML.
- The duality is functorial, extending Stone's correspondence to the orthomodular category.

Key difference from Boolean Stone duality. In the Boolean case, filters and ultrafilters coincide for the purpose of duality (every filter extends to an ultrafilter, and the ultrafilter space suffices). In the OML case, non-maximal filters carry essential information — the dual space must include *all* filters. This is because the failure of distributivity means that the ultrafilter extension theorem fails.

What this provides for the OML extension problem. The duality makes the extension question precise: given a state s on $\mathfrak{A}$, we now have a well-defined target space $\mathcal{S}_0(\mathfrak{A})$ on which to ask whether s extends to a σ-additive measure. In the Boolean case, $\mathcal{S}_0(\mathfrak{A})$ is the classical Stone space and the answer is yes (Carathéodory / Paper I's stone_measure_exists).

What this does NOT do. The paper is purely algebraic/ topological — it establishes the duality but does not address measure theory on $\mathcal{S}_0(\mathfrak{A})$. No states, no measures, no extension theorems. The measure-theoretic question is entirely open.

Questions.

1. What are the topological properties of $\mathcal{S}_0(\mathfrak{A})$ for concrete OMLs (e.g., $L(\mathcal{H})$, Greechie lattices, MO_2)? Is it compact? Hausdorff? Zero-dimensional?
2. Does Gleason's theorem for $L(\mathcal{H})$ correspond to a natural measure-extension result on $\mathcal{S}_0(L(\mathcal{H}))$, or do the two live in different categories?
3. Can the "all filters" structure of $\mathcal{S}_0$ explain why OML measure extension is harder than Boolean measure extension — the larger dual space makes it harder for measures to be strictly positive?
4. Is there a Riesz-type representation connecting states on $\mathfrak{A}$ to measures on $\mathcal{S}_0(\mathfrak{A})$?

Part II
The finite/countable boundary

Where CE lives: above finitary consistency (Deligne, Gödel completeness), below ω_1-completeness (Karp, Espíndola). The boundary is precisely located from three angles:

- **Philosophical:** Howson — Cox's axioms produce finite additivity only; no "natural extension" to $L_{\omega_1,\omega}$ gives σ-additivity.
- **Logical:** Frot — Deligne (enough points for coherent topoi) = finitary completeness = consistency. σ-additivity escapes coherence.
- **Topos-theoretic:** Espíndola — ω_1-coherent completeness exists but Property T on the CE site collapses to classical conditions. The machinery doesn't produce new content at this site.

3 Howson — "De Finetti, Countable Additivity, Consistency and Coherence" (2008)

READ (2026-05-17). Verdict: identifies the same gap (finite additivity vs. σ-additivity) but does not formalize it. TERMINOLOGICAL WARNING: Howson uses "consistency" and "coherence" as synonyms for de Finetti's criterion (finite additivity / no Dutch book). σ-additivity sits above both. This is opposite to

the programme's usage, where consistency = finite additivity and coherence = the stronger condition including σ-additivity.

Howson clarifies de Finetti's position on countable additivity and shows it is internally consistent. Key findings:

The mistranslation. De Finetti's Italian *coerenza* = "consistency" (*consistenza*). His 1972 book uniformly uses "consistency." The English translators (Kyburg) imposed "coherence" against de Finetti's own usage (p. 3). The programme's terminology aligns with de Finetti's original, not the mistranslation.

Consistency = finite additivity, precisely. De Finetti's criterion: P is consistent iff no finite combination of bets at the P-rates guarantees a loss. Characterizes exactly the finitely additive probability functions. Two properties parallel first-order logic: *absoluteness* (consistency depends only on $P[E]$, not on which algebra E sits in) and *compactness* (inconsistency detected by a finite subset). Both fail for σ-additivity (pp. 8–9).

De Finetti's argument against the Dutch Book for σ-additivity (§§3–4). The infinite fair lottery ($p_n = 0$ for all n) is Dutch-Bookable only if postulate (A) — "a finite sum of fair bets is fair" — is extended to countable sums. De Finetti: this extension *entails* σ-additivity, making the Dutch Book argument circular.

Cox → finite additivity only (§6, p. 19). Cox's three scale-free axioms + propositional logic → finitely additive probability. "Cox's result does not extend to endorsing countable additivity. There are infinitary propositional languages… $L_{\omega_1,\omega}$… but there is no natural extension of Cox's proof which would exploit that additional structure."

The open problem (p. 17). "Something is nevertheless missing, and that is a type of completeness theorem telling us that the rules of probability extend to finite but not countable additivity." This is precisely the gap that Frot and Espíndola formalize: Deligne (finitary completeness) = consistency; CE requires ω_1-completeness.

Contact with programme. Howson (2008) is the primary prior-art citation for the companion note. He identifies the gap (finite additivity has compactness, σ-additivity doesn't) and states the open problem (p. 17: "missing completeness theorem"). The companion note makes this precise via ultraproduct proof. The Frot/Espíndola boundary is the formal version of Howson's philosophical observation: Deligne (finitary completeness) = consistency; CE requires something beyond (ω_1-completeness). Cox's axioms are the right starting material for the programme: they produce finite additivity from structural principles; σ-additivity requires something extra that is not grounded the same way.

Five-traditions unification: DEAD (2026-05-18). Abramsky's contextuality obstruction is compact (finite covers); the σ-additivity gap is non-compact (invisible to finite tests). Category error. Removing Abramsky reduces the thesis to Howson (2008) with fancier coordinates. No new theorem.

Source: *BJPS* 59(1):1–23.

4 Frot — "Gödel's Completeness and Deligne's Theorem" (2013)

READ (2026-05-17). Verdict: confirms the dictionary, places CE.

Proves Deligne's theorem ("coherent topoi have enough points") is equivalent to Gödel's completeness for first-order logic. The dictionary: models = points of the topos (geometric morphisms $\mathcal{E} \to \mathbf{Set}$); consistency = non-triviality; coherent theory = finitary axioms = coherent topos.

Original question: Is CE's failure exactly "the classifying topos lacks enough points"? **Answer: Not exactly.** CE is not expressible in coherent (finitary) logic (confirming Łoś), so Deligne's theorem does not apply. CE lives *above* the coherent level — the guarantee of enough points stops before reaching σ-additivity. The right framing: CE requires a completeness theorem for the *infinitary* fragment where σ-additivity lives.

What connects. Finite additivity IS coherent — Biesel's result (sheaf for finite covers) is the measure-theoretic face of Deligne. σ-additivity requires countable operations, escaping coherence. Frot makes this boundary precise: it is the boundary between finitary and infinitary logic, equivalently between coherent and non-coherent toposes.

Source: arXiv:1309.0389.

5 Espíndola — "Infinitary Generalizations of Deligne's Completeness Theorem" (2017; published *JSL* 85(3), 2020)

READ (2026-05-17). Verdict: places CE at $\kappa = \omega_1$; identifies Property T as the structural condition.

Generalizes Deligne to κ-geometric logic (conjunctions and disjunctions of $< \kappa$ formulas, existential quantification over $< \kappa$ variables). Key results:

- **Theorem 2.11.** If κ is regular with $\kappa^{<\kappa} = \kappa$, κ-geometric theories of cardinality $\leq \kappa$ are complete w.r.t. **Set**-valued models.
- **Corollary 3.1.** Under same hypothesis, every κ-separable topos has enough κ-points.
- **Theorem 3.4.** If κ is weakly compact, every κ-coherent topos has enough κ-points.
- **Corollary 3.3.** κ-coherent theories of cardinality $\leq \kappa$ are complete (= Karp's completeness for $\mathcal{L}_{\kappa,\kappa}$).

Property T (Definition 2.2): an exactness condition on the category — every proper diagram (tree with bar) is completely proper. Combines strong distributivity + dependent choice. At $\kappa = \omega$, property T is automatic (Remark 2.7), recovering classical Deligne. At $\kappa > \omega$ it is a genuine restriction. Rule T (p. 1150) is the syntactic counterpart.

Where CE lives. σ-additivity uses countable universal quantification and countable conjunction $\to$ expressible in $L_{\omega_1,\omega}$ $\to$ relevant level is $\kappa = \omega_1$. At this level:

- Theorem 2.11 applies under CH ($\omega_1^{<\omega_1} = \omega_1$).
- Theorem 3.4 requires ω_1 weakly compact (large cardinal axiom).

Reduction check (2026-05-17). Property T at $\kappa = \omega_1$ on the site of finite Boolean subalgebras reduces to σ-completeness of the underlying algebra + uniqueness of the Carathéodory extension. Both classical, both already assumed in any CE setup. No gap for a theorem. Fifth instance of the pattern: CE-adjacent seeds die because the topos/sheaf/logical machinery, applied to the specific site where CE lives, restates classical conditions.

Status. The *coordinates* are genuine (CE lives at $\kappa = \omega_1$ in Espíndola's hierarchy, above Deligne but within Karp). Useful for exposition (Paper I) and for the Caramello reading direction (specifies what her machinery would need to do). Does not produce a seed. The Caramello direction remains open — her Morita equivalence approach is structurally different from "apply Property T to this site."

Source: arXiv:1709.01967.

6 Caramello — *Theories, Sites, Toposes* (2018)

READ (2026-05-17). Verdict: the bridge technique cannot see CE. σ-additivity is not expressible in finitary geometric logic, so it cannot appear as a quotient in

Caramello's framework.

Read Chapters 1, 3–10. The book develops the "bridge technique": given a geometric theory $\mathbb{T}$ of presheaf type with canonical Morita equivalence

$$\mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{op}, J_{\text{at}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}),$$

transfer properties between the syntactic site $(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'})$ and the semantic site $(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{op}, J_{\text{at}})$ to obtain new results.

The discriminating constraint. A "quotient" of a geometric theory $\mathbb{T}$ means adding a *geometric* sequent — built from finite conjunctions, small disjunctions, existential quantification. σ-additivity requires *countable* universal quantification and countable conjunction: it is not a geometric axiom. So σ-additivity literally cannot be expressed as a quotient of any geometric theory in Caramello's framework. The bridge technique cannot see the finite/countable boundary where CE lives — for the same structural reason that Deligne/Frot cannot: finitary expressiveness.

The infinitary escape. At $\kappa = \omega_1$ (Espíndola's generalization), σ-additivity becomes expressible in κ-geometric logic. But we already established that Property T at ω_1 on the CE site reduces to classical conditions. The bridge either can't see CE (finitary) or reduces it to known content (infinitary).

What the book does provide:

- Theorem 3.2.5 (Duality): bijection between quotients of $\mathbb{T}$ and subtoposes of $\mathbf{Set}[\mathbb{T}]$.
- Chapter 4: coHeyting algebra on subtoposes, Booleanization, DeMorganization.
- Chapter 6 (§6.1.1): canonical Morita equivalences for presheaf-type theories.
- Chapter 10: applications — Fraïssé's theorem topos-theoretically, Galois representations.
- p. 337(b): the theory of decidable Boolean algebras is of presheaf type; f.p. models = finite Boolean algebras; homogeneous models = atomless Boolean algebras. But this is the *algebraic* theory — no measures, charges, or σ-additivity.

No measure-theoretic applications. Caramello has no published work applying the bridge technique to measure theory, valuations, or probability.

Contact with programme: None directly. The bridge technique operates within finitary geometric logic, which cannot express σ-additivity. Sixth instance of the pattern: CE is invisible to every finitary framework tried (Zafiris, Biesel, Frot/Deligne, Espíndola-reduced, Caramello).

Source: Oxford University Press, 2018. ISBN 978-0-19-875891-4.

7 Biesel — "Sheaves of Probability" (2024)

READ (2026-05-17). Verdict: confirms dictionary translation.

Proves measures form a sheaf for finite covers (Thm 8), probability measures likewise (Thm 11). Countable covers explicitly excluded (Remark 7: uniform measures on $\{1, \ldots, n\}$ fail to glue to $\mathbb{N}$). Section 5 states explicitly that the results hold for merely finitely additive charges. For countable covers, σ-finiteness is needed — standard measure theory, no hidden structure.

Original question: Is the step from finite to countable covers exactly the CE boundary? **Answer: Yes, but tautologically.** Finitely additive = sheaf for finite covers; σ-additive = sheaf for countable covers. This is a restatement, not a characterization.

Useful residue: Thms 8 and 11 are clean citations for Paper I's CE discussion. Ref [2] (Ross 2012, "All roads lead to violations of countable additivity," *Phil. Studies*) relevant for Howson/de Finetti direction.

Source: arXiv:2401.01968.

8 Vickers — *Topology via Logic* (1989)

READ (2026-05-17). Verdict: provides the framework (locales, sublocales, Stone duality) but not the application to measures. The question remains open — requires later literature.

The book is a domain theory / logic textbook. It develops:

- Locales (§5.4): points = frame homomorphisms to **2**; spatialization/localization functors.
- Sublocales (§6.2): = nuclei on frames = frame quotients. The sublocale of "realized points" (principal ultrafilters) is well-defined in this framework.
- Spectral algebraic locales (§§9.2–9.3): coherent presentations, compact coprime opens, Stone duality.
- Stone spaces (§9.5): spectral + Hausdorff $\leftrightarrow$ Boolean algebra. Clopens = elements of the Boolean algebra. Booleanization functor. Patch topology.
- ω-algebraicity (§9.4): second countability, ω-chain completions. The finite/countable boundary appears here — finite locales are spectral algebraic (Prop. 9.4.1); second countable $\leftrightarrow$ ω-algebraic (Thm. 9.4.6).

What the book does NOT provide:

- No theory of valuations/measures on locales.
- No discussion of σ-additivity or CE in locale-theoretic terms.
- No characterization of "a valuation is supported on the sublocale of principal ultrafilters."

Contact with programme. The framework is now in hand. The open question is whether Vickers's later localic measure theory (2011) provides the valuation-on-locale machinery needed to formulate "μ is supported on the sublocale of principal ultrafilters" as a frame-internal condition. This would be the next reading direction if the CE question is revisited.

Source: Cambridge University Press, 1989. ISBN 0 521 36062 5.

9 Vickers — "A localic theory of lower and upper integrals" (2008)

Question while reading: Can "a valuation is supported on the sublocale of principal ultrafilters" be formulated as a frame-internal condition on a localic valuation? Does σ-additivity of the valuation emerge as a locale-theoretic property (e.g., preservation of directed joins, or a continuity condition on the valuation map)?

Why. Vickers (1989) provides the framework (locales, sublocales, Stone duality) but not the application to measures. This later work develops valuations on locales constructively. The programme's question — is CE a frame-internal property? — requires this machinery. If the support condition "μ(sublocale of realized points) = 1" can be expressed without reference to points, it would be the non-circular characterization of coherence.

Contact with programme: CE characterization directly. This is the actual target that Vickers (1989) pointed toward.

Source: *Math. Logic Quarterly* 54(1), 109–123.

Part III
Paper II adjacent

10 Rédei & Summers — "Quantum Probability Theory" (2006)

READ (2026-05-17). Verdict: review paper, Paper II reference.

Reviews quantum probability in the von Neumann algebra framework. Classical probability = abelian von Neumann algebra; quantum = nonabelian. Normal states (= σ-additive) characterized by continuity. The type I/II/III classification controls which probability-theoretic phenomena appear.

Original question: What is the obstruction to extending states across non-Boolean joins? **Answer:** Noncommutativity itself. The type classification refines this but no single algebraic condition beyond "non-distributive + dim ≥ 3" is isolated.

Useful for Paper II: clean reference for the parallel between classical and quantum probability in the operator algebra framework.

Source: arXiv:quant-ph/0601158.

11 Döring & Isham — "What is a Thing?" (2008)

READ (2026-05-17). Verdict: Paper II reference. Daseinisation is descent-adjacent but not CE-relevant.

The topos is $\mathbf{Sets}^{\mathcal{V}(\mathcal{H})^{op}}$: presheaves on the poset $\mathcal{V}(\mathcal{H})$ of commutative subalgebras of $B(\mathcal{H})$. The spectral presheaf $\underline{\Sigma}$ assigns to each Boolean context V its Gelfand spectrum; it has no global elements (= Kochen–Specker). Daseinisation (§5.2–5.3) maps a projection $\hat{P}$ to a sub-object of $\underline{\Sigma}$ by approximating $\hat{P}$ within each Boolean context.

Original question: Does daseinisation give descent from Stone space to realization? **Answer:** Structurally adjacent but different. Daseinisation goes *up* (approximating non-Boolean by Boolean); descent goes *down* (Stone space to realized points). The failure of global sections IS Kochen–Specker, reformulated topos-theoretically — the non-Boolean analogue of CE failure, but with a different obstruction mechanism (noncommutativity vs. non-σ-additivity).

Contact with programme: Paper II directly. Not CE-relevant.

Source: arXiv:0803.0417.

12 Epperson & Zafiris — *Foundations of Relational Realism* (2013)

READ (2026-05-17). Verdict: does not address CE.

Read Chapters 6, 8, 9, 10 of the book and Zafiris (2006) *J. Math. Phys.* 47, 092103 (the paper that puts measures on the site). The Grothendieck topology J (epimorphic families) does not grade covers by cardinality — finite, countable, and uncountable covers satisfy J uniformly. The finite/countable boundary where CE lives is invisible. States are finitely additive throughout; σ-completeness is an axiom, not a sheaf condition. No cohomology is computed; the paper proves a *reconstruction* theorem (counit is iso), not an obstruction theorem. The H^1 unification with Abramsky–Brandenburger does not appear.

Original question: Does Zafiris's Boolean localization give a sheaf-theoretic formulation of when local states glue to a global σ-additive state? **Answer: No.**

Part IV
Parked directions

Dzhafarov–Kujala, Contextuality by Default — PARKED

Primary: E. N. Dzhafarov and J. V. Kujala, "Contextuality-by-Default 2.0," *Lecture Notes in Computer Science* 10106 (2017), 16–32. See also Dzhafarov and Kujala, "Context-content systems of random variables," *Philosophical Transactions of the Royal Society A* 375 (2017), 20160389.

Status: PARKED (2026-05-19). Read in full (LNCS chapter, pp. 16–32). CbD starts from the position that random variables measured in different contexts *lack a joint distribution primitively.* A joint distribution (coupling) is an achievement, not a given. Contextuality is then the failure of a maximally connected coupling to exist. The framework uses context-content systems R_q^c (double-indexed random variables), multimaximal couplings, and a quasi-coupling/total-variation measure of contextuality.

Discriminating question (as posed): Does CbD's coupling-existence criterion have an analogue for the FA $\to$ σA extension problem?

Verdict: the question was malformed. The coupling problem in CbD is the *Kolmogorov extension problem*: given consistent finite-dimensional marginals, does a joint distribution exist? The FA/σA extension problem is different: given a finitely additive charge on a Boolean algebra, does it extend to a σ-additive measure on the σ-completion? These are distinct extension problems with different obstructions. CbD's "refinement contexts" are finite marginal distributions that are already σ-additive; the coupling asks whether they paste into a global σ-additive joint. Our problem asks whether a single charge on a single algebra can be promoted from finite to countable additivity — no pasting of marginals is involved.

Additionally, CbD contextuality is *compact*: it is witnessed by finite subsystems (a finite set of contexts suffices to detect non-existence of a coupling). The FA/σA gap is *non-compact*: no finite test can distinguish a σ-additive measure from a purely finitely additive charge (this is precisely the non-first-order-axiomatizability established in the companion note). The compact/non-compact mismatch is the same structural reason that Abramsky–Brandenburger sheaf contextuality does not transfer.

The Yosida–Hewitt echo (quasi-couplings decomposing into signed and positive parts, paralleling $\mu = \mu_c + \mu_p$) is vocabulary, not theorem. No new content about the FA/σA boundary.

Ellerman, Partition Logic and Information — PARKED

Primary: D. Ellerman, "An Introduction to Partition Logic," *Logic Journal of the IGPL* 22 (2014), 94–125. See also Ellerman, "The Logic of Partitions," *Review of Symbolic Logic* 3 (2010), 287–350; and *New Foundations for Information Theory*, Springer (2021).

Status: PARKED (2026-05-19). Read in full. Ellerman builds a complete dual logic where partitions (equivalence relations) play the role that subsets play in Boolean logic. Distinctions ("dits") dualize elements; the partition lattice $\Pi(U)$ is non-distributive; each partition π generates a Boolean core $\mathcal{B}_\pi \cong \mathcal{P}(\pi_{ns})$. "Logical entropy" $h(\pi) = \sum p_i(1 - p_i)$ is the normalized dit-counting measure.

Verdict on discriminating question: The dit/bit translation *is* relabeling entropy. Logical entropy $h(\pi) = 1 - \sum p_i^2 = 1 - \mathrm{Coll}_\mu(\mathcal{G})$, i.e., one minus the collision probability already used in the programme's observational-resolution notes. The partition refinement order recovers the cylinder-algebra filtration. The paper stays entirely in the finite setting — no measures, no charges, no limit behavior, no extension problems. No theorem in partition logic translates to new content about the FA/σA boundary.

OML re-scoping (rejected): The non-distributivity of $\Pi(U)$ does not connect to the OML extension problem. Partition lattices are semimodular (geometric) lattices, not orthomodular lattices: the equivalence-relation complement is not an orthocomplement, and the failure of distributivity in $\Pi(U)$ is of a different structural kind than in $L(\mathcal{H})$ or a general OML. Wrong kind of non-distributivity; no transfer.

Walley, Imprecise Probability — PARKED

Primary: P. Walley, *Statistical Reasoning with Imprecise Probabilities*, Chapman & Hall (1991). For a concise treatment, see M. C. M. Troffaes and G. de Cooman, *Lower Previsions*, Wiley (2014).

Walley takes as primitive a *set* of probability measures (a credal set), not a single measure. Sharp (precise) probability is the special case where the credal set is a singleton. The framework has a natural hierarchy: upper/lower previsions $\to$ coherent previsions $\to$ precise previsions $\to$ σ-additive previsions. Each inclusion is strict. The theory of "natural extension" — the smallest coherent prevision dominating a given assessment — is formally analogous to the problem of extending a finitely additive charge.

Assessment (read §§6.8–6.9, Appendix K): PARKED — discriminating question was malformed.

The discriminating question conflated two distinct axes in Walley's hierarchy. In his framework, the credal set $\mathcal{M}(\underline{P})$ is the set of linear previsions (finitely additive probabilities) dominating a lower prevision $\underline{P}$. Singleton $\mathcal{M}(\underline{P})$ means *precision* (sharp probability), not σ-additivity. σ-additivity corresponds to *conglomerability*: a prevision P is $\mathcal{B}$-conglomerable if conditional and unconditional previsions cohere across a partition $\mathcal{B}$. Full conglomerability (for all partitions) is equivalent to countable additivity for precise previsions (Theorem 6.9.2, crediting Schervish–Seidenfeld–Kadane 1984).

The framework thus has *two* independent axes: (i) precision (sharp vs. imprecise), (ii) conglomerability (σ-additive vs. merely finitely additive). The discriminating question asked about axis (i) as if it were axis (ii).

The load-bearing mathematics is SSK (1984): conglomerability $\leftrightarrow$ countable additivity, already in the programme via Howson (2008) and the companion note. Walley's original contribution is normative (decision-theoretic interpretation of the FA/σA hierarchy as a rationality axiom, his P8), not mathematical (no new extension theorem). The "natural extension" machinery (§3.4) is the lower-envelope construction, formally dual to the Yosida–Hewitt decomposition but not producing new structural results about when extensions are σ-additive.

W-coherence (Appendix K), a weakening of coherence that drops the conglomerative principle, always admits W-coherent extensions to larger domains. This is a rationality-axiom distinction, not a structural one: it does not address whether a given finitely additive charge has a σ-additive extension.

Cluster: Same as Howson (2008). The imprecise-probability framework redescribes the FA/σA hierarchy in decision-theoretic language without adding new theorems about when σ-additive extension exists. Vocabulary, not mathematics.

Abramsky–Brandenburger, Sheaf-Theoretic Formalism — PARKED

Primary: S. Abramsky and A. Brandenburger, "The Sheaf-Theoretic Structure of Non-Locality and Contextuality," *New Journal of Physics* 13 (2011), 113036.

Re-scoped. The contextuality application was previously killed: the contextuality obstruction is compact (witnessed by finite covers), while the σ-additivity gap is non-compact (invisible to any finite test). These are different phenomena.

What survives is the *formalism*: a presheaf of local sections (compatible local data), a sheaf condition (existence of a global section), and a cohomological obstruction (H^1 or higher) measuring the failure. The question is whether this formal apparatus can be applied to the Yosida-Hewitt decomposition, where the "local sections" are the restrictions of a finitely additive charge to finite subalgebras and the "global section" is a σ-additive extension.

PARKED (2026-05-19). Read in full. The formalism does not apply, and the kill is stronger than the compact/non-compact mismatch alone.

The cohomology is trivially zero on the natural target. Stone spaces are totally disconnected compact Hausdorff; their Čech cohomology with constant coefficients vanishes in all positive degrees. Applied to $\mathrm{St}(B)$ for a Boolean algebra B, the AB obstruction theory gives $H^1 = 0$ unconditionally. The purely finitely additive component μ_p *cannot* appear as a cohomological obstruction because there is no cohomology to obstruct.

The underlying mismatch. The AB framework is combinatorial: finite measurement covers (anti-chain families on a finite set of observables), finite incidence matrices, LP duality for signed measures (Theorem 5.5/5.9). Čech cohomology in the AB sense operates on finite simplicial complexes built from finite measurement scenarios. The signed-measure result (every no-signaling model has a signed global section) is LP duality on finite matrices, not the Yosida-Hewitt decomposition. Extending the formalism to countable limits would require pro-sheaf or ind-sheaf machinery that does not exist in the AB literature and would constitute new construction, not application.

Verdict: The compact/non-compact mismatch (already identified when killing the "five traditions" thread) is confirmed, and the stronger cohomological vanishing makes rescue impossible without rebuilding the formalism. No programme contact.

Sambin, Positive Topology — PARKED

Primary: G. Sambin, "Some points in formal topology," *Theoretical Computer Science* 305 (2003), 347–408. See also Sambin, *Positive Topology: Completing the Foundational Trilogy* (book in preparation, drafts circulating since 2020).

Status: PARKED (2026-05-19). Read the 2003 TCS paper in full. The framework is purely topological/lattice-theoretic with no measure-theoretic content.

What Sambin provides. A basic pair $(X, \Vdash, S)$ relates points and observables through a forcing relation. A formal topology $(S, A, \triangleleft, \kappa)$ has: a cover $\triangleleft$ (inductive, from an axiom set A) playing the role of topology; and a positivity predicate κ (co-inductive, from a compatibility set J) replacing the classical complement. Formal points are derived: a filter $\alpha \subseteq S$ must be inhabited, convergent, split covers, and enter κ.

Why the primitives are the wrong shape. The positivity predicate is co-inductive — it is the constructive replacement for topological inhabitedness (an open is "positive" if it contains a formal point). This is *not* a σ-additivity test. The Yosida-Hewitt decomposition requires σ-completion machinery (countable limits, continuity from above) that lives outside formal topology. Even if one added measures to Sambin's framework, the positivity predicate would not characterize the σ-additive component: κ detects topological non-emptiness, not measure-theoretic continuity at countable joins. The "completion" in Sambin's sense is spatial completion (adding formal points to make a locale spatial), not measure extension.

Verdict: Redescribes Stone duality constructively without engaging measure theory. The structural disconnect is not just "no measures present" but "the available primitives (cover + positivity) are the wrong type to express σ-additivity." No programme contact.

Gisin (2021) — PARKED

Primary: N. Gisin, "Indeterminism in Physics, Classical Chaos and Bohmian Mechanics," *Erkenntnis* 86 (2021), 1553–1590.

Status: PARKED (2026-05-19). Read in full. Physics/philosophy paper with zero measure-theoretic content. Proposes "finite-information numbers" (determined prefix + undetermined tail); shows classical mechanics becomes non-deterministic but empirically equivalent. No probability, no charges, no Boolean algebras, no extension problems. The philosophical resonance (infinite precision is unphysical / σ-additivity requires infinite verification) is real but shallow: Gisin's finite-information numbers are a physical proposal, not a constructive real-number system in the Bishop/Brouwer sense, and the paper develops no measure theory on such objects. No formal contact with the FA/σA question.

Linnebo–Shapiro (2019) — PARKED

Primary: Ø. Linnebo and S. Shapiro, "Actual and Potential Infinity," *Noûs* 53 (2019), 160–191.

Status: PARKED (2026-05-19). Read in full. The paper develops modal explication of potential infinity via S4.2: liberal potentialism preserves classical logic (potentialist translation $\forall \to \Box\forall$, $\exists \to \Diamond\exists$); strict potentialism yields intuitionistic logic (realizability interpretation, Heyting arithmetic). This concerns quantification over potentially infinite domains, not measure extension. σ-additivity is a property of charges on algebras, not a quantifier pattern; the FA/σA gap is already known to be non-FO-axiomatizable (Tarski); switching to intuitionistic logic does not change which Boolean algebras admit σ-additive measures (a set-theoretic/topological question, not proof-theoretic).

Note: Hamkins's *set-theoretic* potentialism (forcing extensions of the universe) is a separate thread and should not be conflated with Linnebo–Shapiro's mathematical-domain potentialism. The set-theoretic version is directly relevant to Strategy D, where the question "does a non-σ-complete non-atomic measure-free Boolean algebra exist?" may have different truth values in different forcing extensions. If Strategy D is pursued, Hamkins is the right contact point — not Linnebo–Shapiro.

How to use this document

When reading, annotate in the margins or on a separate sheet:

1. What theorem or definition surprised you.
2. What connected to the programme.
3. What contradicted the knowledge map.
4. What suggested a concrete Phase 1 seed note.

When a reading direction produces a concrete claim worth auditing, extract it to `notes/unsorted/` and run Phase 2.